

Real Time API V1

By Bohdan MukhaApr 09, 2026

Real-time API via AMQP protocol and with the help of Azure Service Bus service implemented. Rythmos allows clients to listen to a bunch of different events under one topic, where topic name will be `api/event/$communityId`

Rythmos will provide connection string/access keys for each queue for each community.

All messages are covered in envelopes, with specified event type and event object.

Azure Service Bus on the Rythmos side is distributed and has high availability infrastructure, also it is important to ensure that all events are delivered in case of deployments on client sides, so all messages that were not marked as completed will stay in the queue for further processing.

Location event object

```
1 {
2   "type": "location",
3   "event": {
4     "floorId": "95",
5     "latitude": 39.70676840649414,
6     "longitude": -104.9226303100586,
7     "communityId": "15",
8     "deviceId": "19A0881286",
9     "timeInUnixTicks": 1725014585,
10    "nearestAnchorId": "8652366",
11    "location": "Apartment 315"
12  }
13 }
```

Alert event object

Base alert event

```
1 {
2   "type": "alert",
3   "event": {
4     "eventType": 11, // see enum at the bottom of this doc
5     "communityId": "15",
6     "deviceId": "471549",
7     "timeInUnixTicks": 1724949623660,
8     "nearestAnchorId": null,
9     "location": "1st Floor Main Entrance Door"
10  }
11 }
```

When a staff member accepts or deactivates an alert from the Rythmos backend or apps, the alert event carries an additional `data.staffId` field containing the internal Rythmos user id of the staff member who performed the action. To resolve `staffId` to a full staff profile (name, role, username, facility), use the `GET /api/v1/staff` endpoint on the [REST API V1](#).

Two cases carry `data.staffId`:

- Deactivation by staff** — existing deactivation events (enum values `2001-2026`, suffix `DeactivationFromRythmos`) will include `data.staffId` when the facility is enrolled.
- Accept by staff** — new event emitted when a staff member taps “Accept” on an incoming alert. Uses the new enum values `3001-3026` (suffix `AcceptedByStaff`). The accept enum mirrors the deactivation enum one-for-one with a `+1000` offset (for example, `ButtonPressDeactivationFromRythmos = 2001` ↔ `ButtonPressAcceptedByStaff = 3001`).

Consumers distinguish accept from deactivate by the `eventType` range:

- `2001-2026` → deactivation by staff
- `3001-3026` → accept by staff

Both the accept and deactivation events share the same `id` as the original activation (the pendant event id prefixed with `P`, for example `"P1234567"`), so clients can correlate a full alert lifecycle: activation → accept → deactivation.

Example accept event

```
1 {
2   "type": "alert",
3   "event": {
4     "id": "P1234567",
5     "eventType": 3001,
6     "communityId": "95",
7     "deviceId": "SN-A1B2C3D4",
8     "timeInUnixTicks": 1775999312000,
9     "nearestAnchorId": "10042",
10    "location": "Apt 204 - Living Room",
11    "roomId": "15",
12    "data": {
13      "rythmosResidentId": 789,
14      "staffId": 42
15    }
16  }
17 }
```

Example deactivation event with staff info

```
1 {
2   "type": "alert",
3   "event": {
4     "id": "P1234567",
5     "eventType": 2001,
6     "communityId": "95",
7     "deviceId": "SN-A1B2C3D4",
8     "timeInUnixTicks": 1775999400000,
9     "nearestAnchorId": "10042",
10    "location": "Apt 204 - Living Room",
11    "roomId": "15",
12    "data": {
13      "rythmosResidentId": 789,
14      "staffId": 42
15    }
16  }
17 }
```

Notes

- `data.staffId` is populated only for facilities explicitly enrolled in the staff id stream. Facilities that are not enrolled continue to receive events in the existing shape, without a `data.staffId` field and without any accept events.
- Consumers that don't know about `data.staffId` can safely ignore the field — it is additive, not breaking.
- Accept events (`3001-3026`) are only emitted for enrolled facilities. Consumers that don't recognize these enum values should treat them as unknown event types and ignore them.
- The same `id` is shared across the activation, accept, and deactivation events for a single alert, so clients can use it as the correlation key.

Example of integration (Java)

This is simple code for listening

```
1 import com.azure.messaging.servicebus.*;
2
3 public class ServiceBusQueueReceiverWithProcessor {
4     private static final String CONNECTION_STRING = "we provide"; // static credential
5     private static final String QUEUE_NAME = "api/events/100"; // Replace with your community id
6
7     public static void main(String[] args) {
8         ServiceBusProcessorClient processorClient = new ServiceBusClientBuilder()
9             .connectionString(CONNECTION_STRING)
10            .processor()
11            .queueName(QUEUE_NAME)
12            .processMessage(ServiceBusQueueReceiverWithProcessor::processMessage)
13            .processError(ServiceBusQueueReceiverWithProcessor::processError)
14            .buildProcessorClient();
15
16        processorClient.start();
17
18        System.out.println("Listening for messages on the queue...");
19
20        Runtime.getRuntime().addShutdownHook(new Thread(() -> {
21            System.out.println("Stopping processor client...");
22            processorClient.close();
23        }));
24    }
25
26    private static void processMessage(ServiceBusReceivedMessageContext context) {
27        ServiceBusReceivedMessage message = context.getMessage();
28
29        System.out.println("Received message with ID: " + message.getMessageId());
30        System.out.println("Message Body: " + new String(message.getBody().toBytes()));
31        // Add your client logic about parsing different types of messages
32
33        context.complete(); // Mark the message as processed successfully
34    }
35
36    private static void processError(ServiceBusErrorContext context) {
37        System.out.println("Error occurred: " + context.getException().getMessage());
38        if (context.getException() != null) {
39            context.getException().printStackTrace();
40        }
41    }
42 }
```

Enums used in the Real-Time API

```
1 enum AlertType (or EventType)
2 {
3     Unknown = 0,
4     ButtonPressActivation = 1,
5     ButtonPressDeactivation = 2,
6     SoftFallActivation = 3,
7     HardFallActivation = 4,
8     FallDeactivation = 5,
9     WanderActivation = 6,
10    PullCordActivation = 7,
11    PullCordDeactivation = 8,
12    WindowSensorOpen = 9,
13    WindowSensorClose = 10,
14    DoorSensorOpen = 11,
15    DoorSensorClose = 12,
16    SmokeDetectorActivation = 13,
17    SmokeDetectorDeactivation = 14,
18    DoorBellActivation = 15,
19    DoorBellDeactivation = 16,
20    MotionSensorActivation = 17,
21    MotionSensorDeactivation = 18,
22    PresenceDetectionSensorNoDetected = 19,
23    PresenceDetectionSensorStart = 20,
24    PresenceDetectionSensorEnd = 21,
25    AlexaHelpRequestActivation = 23,
26    LoiteringActivation = 24,
27    LoiteringDeactivation = 25,
28    GeofenceKeepOut = 26,
29    DoorBreached = 28,
30    DoorPropped = 29,
31    DoorOpened = 30,
32    DoorClosed = 31,
33    MeshDisconnect = 32,
34    MeshConnect = 33,
35    GeofenceKeepIn = 34,
36    DoorBreachedAttempt = 36,
37    OutOfBedAlert = 37,
38
39    // Cellular device
40    CellularButtonPressActivation = 1001,
41    CellularButtonPressDeactivation = 1002,
42    CellularSoftFallActivation = 1003,
43    CellularSoftFallDeactivation = 1004,
44    CellularHardFallActivation = 1005,
45    CellularHardFallDeactivation = 1006,
46    CellularWanderActivation = 1007,
47
48    // Deactivation by staff (from Rythmos backend feature or app)
49    ButtonPressDeactivationFromRythmos = 2001,
50    FallDeactivationFromRythmos = 2002,
51    WanderDeactivationFromRythmos = 2003,
52    PullCordDeactivationFromRythmos = 2004,
53    WindowSensorOpenedDeactivationFromRythmos = 2005,
54    WindowSensorClosedDeactivationFromRythmos = 2006,
55    DoorSensorOpenedDeactivationFromRythmos = 2007,
56    DoorSensorClosedDeactivationFromRythmos = 2008,
57    SmokeDetectorDeactivationFromRythmos = 2009,
58    DoorBellDeactivationFromRythmos = 2010,
59    MotionSensorDeactivationFromRythmos = 2011,
60    PresenceDetectionSensorMotionDeactivation = 2012,
61    AlexaHelpRequestDeactivationFromRythmos = 2013,
62    LoiteringDeactivationFromRythmos = 2014,
63    GeofenceDeactivationFromRythmos = 2015,
64    WearableRemovedDeactivationFromRythmos = 2016,
65    DoorBreachedDeactivationFromRythmos = 2017,
66    DoorProppedDeactivationFromRythmos = 2018,
67    DoorOpenedDeactivationFromRythmos = 2019,
68    DoorClosedDeactivationFromRythmos = 2020,
69    IncontinenceAlertDeactivationFromRythmos = 2021,
70    DoorBreachedAttemptDeactivationFromRythmos = 2022,
71    OutOfBedAlertDeactivationFromRythmos = 2023,
72    CellularButtonPressDeactivationFromRythmos = 2024,
73    CellularFallDeactivationFromRythmos = 2025,
74    CellularWanderDeactivationFromRythmos = 2026,
75
76    // Accept by staff — mirrors the deactivation range above with a +1000 offset
77    ButtonPressAcceptedByStaff = 3001,
78    FallAcceptedByStaff = 3002,
79    WanderAcceptedByStaff = 3003,
80    PullCordAcceptedByStaff = 3004,
81    WindowSensorOpenedAcceptedByStaff = 3005,
82    WindowSensorClosedAcceptedByStaff = 3006,
83    DoorSensorOpenedAcceptedByStaff = 3007,
84    DoorSensorClosedAcceptedByStaff = 3008,
85    SmokeDetectorAcceptedByStaff = 3009,
86    DoorBellAcceptedByStaff = 3010,
87    MotionSensorAcceptedByStaff = 3011,
88    PresenceDetectionSensorMotionAcceptedByStaff = 3012,
89    AlexaHelpRequestAcceptedByStaff = 3013,
90    LoiteringAcceptedByStaff = 3014,
91    GeofenceAcceptedByStaff = 3015,
92    WearableRemovedAcceptedByStaff = 3016,
93    DoorBreachedAcceptedByStaff = 3017,
94    DoorProppedAcceptedByStaff = 3018,
95    DoorOpenedAcceptedByStaff = 3019,
96    DoorClosedAcceptedByStaff = 3020,
97    IncontinenceAlertAcceptedByStaff = 3021,
98    DoorBreachedAttemptAcceptedByStaff = 3022,
99    OutOfBedAlertAcceptedByStaff = 3023,
100   CellularButtonPressAcceptedByStaff = 3024,
101   CellularFallAcceptedByStaff = 3025,
102   CellularWanderAcceptedByStaff = 3026
103 }
```